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Supplementary Materials

Table S1. Complementary pollutants: black carbon (BC) and PM2.5 single-pollutant panel regression estimates. The dependent variable is log(asthma ED visit rate per 10,000, children aged 5-17). All specifications include two-way fixed effects with clustered standard errors at the UHF42 level.

Specification	β (BC)	SE	p	N	β (PM2.5)	SE	p	N
Baseline two-way FE	0.169	0.121	0.162	630	0.022	0.025	0.389	630
Excluding 2020-2021	0.109	0.162	0.502	546	0.020	0.030	0.521	546
Two-way FE with borough trends	—	—	—	—	—	—	—	—
Lead test (t+1)	0.052	0.139	0.706	588	-0.007	0.036	0.851	588
First differences	0.359	0.161	0.026	588	0.054	0.041	0.193	588

Table S2. Confounder sensitivity: NO2 first-differences coefficient with and without time-varying poverty rate and racial composition (percentage non-Hispanic Black) as covariates.

Specification	β (NO2)	SE	p	N
FD: NO_2 only	0.0217	0.0109	0.047	588
FD: NO2 + poverty rate	0.0204	0.0105	0.053	588
FD: NO_2 + pct non-Hispanic Black	0.0216	0.0109	0.048	588
FD: NO2 + poverty + race	0.0202	0.0105	0.056	588

* Adding time-varying poverty rate reduced the NO_2 coefficient by approximately 6%. The combined model (poverty + race) shows a 7% reduction.

Table S3. Alternative health outcomes: panel fixed-effects (FE) and first-differences (FD) estimates. The exposure variable is area-weighted NO2 (ppb). All specifications include two-way fixed effects with clustered standard errors.

Outcome	β (FE)	SE	p	β (FD)	SE	p	N
ED rate, age 5-17 (primary)	0.009	0.010	0.33	0.022	0.011	0.046	630
ED rate, age 0-4	0.005	0.010	0.62	0.020	0.011	0.060	630
Hospitalization rate, age 5-17	-0.027	0.012	0.020	-0.030	0.026	0.244	625
Hospitalization rate, age 0-4	0.008	0.012	0.49	0.071	0.025	0.004	623
ED rate, adults 18+	0.022	0.014	0.12	-0.008	0.009	0.399	336
Hospitalization rate, adults 18+	-0.019	0.019	0.32	-0.022	0.014	0.127	336

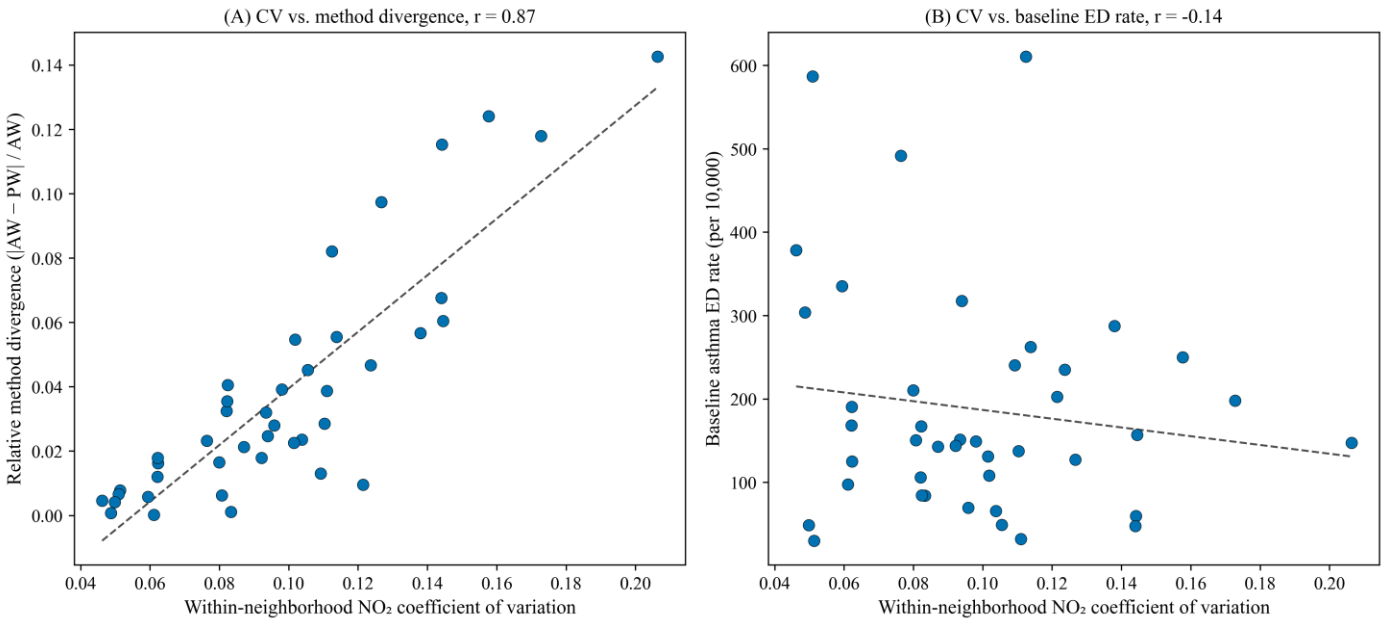

Figure S1. Diagnostic scatterplots: (**A**) within-neighborhood NO_2 coefficient of variation versus relative method divergence (r = 0.87), and (**B**) within-neighborhood NO_2 coefficient of variation versus baseline asthma ED rate (r = -0.14).